

BRIAN A. CLARK

2105 Physical Sciences Complex
University of Maryland, College Park
College Park, MD 20742

Phone: (301)-405-6036
Email: baclark@umd.edu
Website: <https://baclark.physics.umd.edu>
OrcID / inSPIRE: 0000-0003-4089-2245 / Brian.A.Clark.1

ACADEMIC APPOINTMENTS

Assistant Professor of Physics and JSI Junior Fellow University of Maryland & Joint Space Science Institute (JSI), College Park, MD USA	2023-Present
NSF Astronomy and Astrophysics Postdoctoral Fellow Michigan State University, East Lansing, MI USA	2019-2022
NSF Graduate Research Fellow The Ohio State University, Columbus, OH USA	2016-2019
Teaching Assistant The Ohio State University, Columbus, OH USA	2015-2016

EDUCATION

Ph.D. in Physics, The Ohio State University , Columbus, Ohio USA	2014-2019
M.S. in Physics, The Ohio State University , Columbus, Ohio USA	2014-2016
B.A. in Physics (<i>cum laude</i>), Washington University , St. Louis, Missouri USA	2010-2014

FELLOWSHIPS and AWARDS

NSF Astronomy and Astrophysics Postdoctoral Fellowship	2019-2022
Japanese Society for the Promotion of Science Postdoctoral Fellowship	2022
NSF Graduate Research Fellowship	2016-2019
OSU Graduate Enrichment Fellowship	2014-2015
WUSTL Undergraduate Physics Research Fellow	2011
APS Division of Astrophysics Travel Award	2017, 2019
Bunny and Thomas Clark Graduate Scholarship Honorable Mention	2019

EXTERNAL FUNDING

PI of NSF Award PHY-2607903 – \$100,000 “Collaborative Research: WoU-MMA: RNO-G Data Analysis in the U.S. 2026-2029”	2026/09-2027/08
Co-PI of NSF Award PHY-2608108 – \$500,000 “Collaborative Research: WoU-MMA: IceCube Data Analysis in the U.S. 2026-2029”	2026/09-2027/08
PI of NSF Award PHY-2411849 – \$969,640 “WoU-MMA: Extremely High Energy Neutrinos in the IceCube Neutrino Observatory”	2024/09-2027/08
PI of NSF Award PHY-2310125 – \$148,000 “Collaborative Research: WoU-MMA: Ultrahigh Energy Neutrinos with the Radio Neutrino Observatory in Greenland”	2023/09-2025/08
PI of NSF Award AST-1903885 – \$328,750 “Unveiling the Ultra-High Energy Universe with Neutrinos”	2019/08-2022/12

SELECTED PUBLICATIONS (h-index = 46)

By policy, the IceCube, ARA, and RNO-G collaborations list authors in alphabetical order. A full publication list is available at the end of the CV and at my ORCID or INSPIRE-HEP pages.

12. “Observation of In-ice Askaryan Radiation from High-Energy Cosmic Rays”
N. Alden *et al.* for the ARA Collaboration (incl. **B. A. Clark**)
Physical Review Letters 136 151003 (2026). [arXiv:2510.21104]
11. “A search for extremely-high-energy neutrinos and first constraints on the ultra-high-energy cosmic-ray proton fraction with IceCube”
R. Abbasi *et al.* for the IceCube Collaboration (incl. **B. A. Clark**)
Physical Review Letters 135 031001 (2025). [arXiv:2502.01963]
(Named an Editor’s Selection and part of the PRL “2025 Collection of the Year”.)
10. “Instrument design and performance of the first seven stations of RNO-G”
S. Agarwal *et al.* for the RNO-G Collaboration (incl. **B. A. Clark**)
JINST 20 (2025) 04, P04015. [arXiv:2411.12922]
9. “A low-threshold ultrahigh-energy neutrino search with the Askaryan Radio Array”
P. Allison *et al.* for the ARA Collaboration (incl. **B. A. Clark**)
Physical Review D 105, 122006 (2022). [arXiv:2202.07080]
8. “toise: a framework to describe the performance of high-energy neutrino detectors”
J. van Santen, **B. A. Clark**, R. Halliday, S. Hallman, A. Nelles
JINST 17 (2022) T08009. [arXiv:2202.11120]
7. “Simulation and Sensitivity for a phased IceCube-Gen2 deployment”
B. A. Clark, R. Halliday for the IceCube-Gen2 Collaboration
Proc. 37th International Cosmic Ray Conference PoS (ICRC2021)1186. [arXiv:2107.08500]
6. “Sensitivity Studies for the IceCube-Gen2 radio array”
S. Hallmann, **B. A. Clark**, C. Glaser, D. Smith for the IceCube-Gen2 Collaboration
Proc. 37th International Cosmic Ray Conference PoS (ICRC2021)1183. [arXiv:2107.08910]
5. “The IceCube-Gen2 Neutrino Observatory”
B. A. Clark for the IceCube-Gen2 Collaboration
JINST 16 (2021) C10007, Proc. 9th Very Large Volume Neutrino Telescope Workshop (VLVnT-2021). [arXiv:2108.05292]
4. “Constraints on the diffuse flux of ultrahigh energy neutrinos from four years of Askaryan Radio Array Data in two stations”
P. Allison *et al.* for the ARA Collaboration (incl. **B. A. Clark** as corresponding author)
Physical Review D 102, 043021 (2020). [arXiv:1912.00987]
3. “NuRadioMC: Simulating the radio emission of neutrinos from interaction to detector”
C. Glaser *et al.* (incl. **B. A. Clark**)
European Physical Journal C 80, 77 (2020). [arXiv:1906.01670]
2. “Observation of Reconstructable Radio Emission Coincident with an X-Class Solar Flare in the Askaryan Radio Array Prototype Station.”
P. Allison *et al.* for the ARA Collaboration (incl. **B. A. Clark** as corresponding author)
Submitted to Astroparticle Physics. [arXiv:1807.03335]
1. “Analyzing the Data from X-ray Polarimeters with Stokes Parameters.”
F. Kislat, **B. Clark**, M. Bielicke, H. Krawczynski.
Astroparticle Physics 68, 45-51 (2015). [arXiv:1409.6214]

SCIENTIFIC TALKS & POSTERS

Invited Talks

20. Physics and Astronomy Colloquium, James Madison University, Harrisonburg VA	2026/04/02
19. HEP Seminar, Michigan State University, East Lansing MI	2026/03/24
18. APS Mid-Atlantic Meeting, Harrisburg PA	2025/11/15
17. Seminar, Penn State Center for Gravitation and the Cosmos, State College PA	2025/09/29
16. Seminar, High Energy Astrophysics at Goddard Space Flight Center, Greenbelt MD	2025/09/25
15. Seminar, Center for Theory and Computation at UMD, College Park MD	2025/09/24
14. Plenary, Lake Louise Winter Institute, Lake Louise Canada	2025/03/06
13. Physics Colloquium, The Ohio State University, Columbus OH	2024/10/22
12. Plenary, TeV Particle Astrophysics, Chicago IL	2024/08/28
11. Physics Colloquium, University of Virginia, Charlottesville VA	2023/12/01
10. Seminar, Yale University Wright Lab, New Haven CT	2023/05/04
9. Seminar, IIHE at VUB/ULB, Brussels BE	2022/10/14
8. Plenary, 15th Intl. Conf. on Particle Phys and Cosmology (PPC), St. Louis MO	2022/06/09
7. HEP Seminar, Univ. of Maryland, College Park MD	2022/02/23
6. Physics Colloquium, Drexel University, Philadelphia PA	2022/02/17
5. Physics & Astronomy Colloquium, Univ. of Delaware, Newark DE	2022/02/09
4. Physics & Astronomy Colloquium, Univ. of Kansas, Lawrence KS	2021/11/22
3. Plenary, Very Large Volume Neutrino Telescopes 2021	2021/05/19
2. Astronomy Seminar, Michigan State Univ., East Lansing MI	2019/10/23
1. Physics Colloquium, College of Wooster, Wooster OH	2016/10/04

Contributed Talks and Posters

18. Snowmass 2021 Community Summer Study, Seattle, WA	2022/07/19
17. APS April Meeting 2022, New York NY	2022/04/12
16. 20th Annual AAPF Symposium	2022/01/10
15. International Cosmic Ray Conference 2021, Berlin Germany	2021/07/20
14. APS April Meeting 2021	2021/04/19
13. 19th Annual AAPF Symposium	2021/02/09
12. NEUTRINO 2020, Chicago IL	2020/06/21
11. 18th Annual AAPF Symposium at the 235th AAS Meeting, Honolulu HI	2020/01/04
10. OSU CCAPP Seminar, Columbus OH.	2019/07/16
9. APS April Meeting 2019, Denver CO	2019/04/15
8. Ohio Section of the APS Fall 2018 Meeting, Toledo OH	2018/09/29
7. OSU CCAPP Seminar, Columbus OH	2018/05/22
6. APS April Meeting 2018, Columbus OH	2018/04/16
5. TeV Particle Astrophysics, Columbus OH	2017/08/11
4. APS April Meeting 2017, Washington DC	2017/01/31
3. Computing in High Energy Astropart. Phys. Research 2016, Columbus OH.	2016/05/26
2. OSU Physics Summer Seminar Series, Columbus OH	2016/04/23
1. Ohio Section of the APS Spring 2016 Meeting, Dayton OH	2016/04/09

LEADERSHIP and SERVICE

Scientific Leadership

IceCube Collaboration Speakers Committee Chair	2025-Present
IceCube Collaboration Diffuse Working Group Convener	2022-2025
Early Career Scientists Representative, the IceCube Collaboration	January 2021-January 2023
ARA Analysis Coordinator	2020-2023
ARA Operations Coordinator	2018-2019

Professional Activities

Convener of the DPF 2026 Conference	2026
Reviewer in a NSF Astronomy and Astrophysics Panel	2024, 2026
Reviewer in a NASA Peer Review Panel	2021-2022
Peer Reviewer, Journal of Astroparticle Physics	2022-Present
Peer Reviewer, Journal of Instrumentation	2021-Present

University and Departmental Service

Co-Chair, UMD Physics Departmental Colloquium Committee	2023-Present
Member, UMD Physics Departmental Laboratory Committee	2023-Present
Member, UMD Physics Departmental Admissions Committee	2023-Present
Member, OSU Physics Climate and Diversity Committee	January 2017-May 2018
Officer, OSU Physics Graduate Student Council	October 2014-May 2017

MENTORSHIP and ADVISING

Postdoctoral Researchers

Bennett Brinson	2025-Present
Steve Sclafani	2023-2025

PhD Research Students

Taylor St Jean	2024-Present
Emma Tintinger	2025-Present
Aishwarya Vijai	2022-Present

Undergraduate Research Students

Zoe Brunton	2024-Present
Skye Joegriner	Summer 2025
Ariel Khavulya	2026-Present
Nicolas Manic	2026-Present
Rohan Panchwagh	2023-2025
Santiago Sued	2024-Present

Committee Membership

Abigail Bishop (PhD at UW-Madison)	July 2026
Eli Mazrachi (PhD)	March 2024
Zachary Metzler (PhD)	October 2025
Lucas Smith (PhD)	2023-Present
Elijah Willox (PhD)	April 2024
Zelong Zhang (PhD at SBU)	February 2025

TEACHING

The University of Maryland, College Park, MD*Physics 265: Introduction to Scientific Programming* (Spring 2025, Spring 2026)*Physics 273: Introductory Physics: Waves* (Fall 2025, Fall 2026)*Physics 276: Experimental Physics II* (Spring 2023, Spring 2024, Fall 2024)**The Ohio State University**, Columbus, OH*Astronomy 1143: Stars, Galaxies, and Cosmology* (Spring 2016)*Physics 1251: E&M, Optics, and Quantum Mechanics* (Fall 2015)**OUTREACH**

Talk at UMD PHYS 170 “Professional Physics” Seminar	October 2025
Talk at UMD Physics Undergraduate Committee (PUC) Meeting	October 2025
Communication Workshop for UMD Departmental “Toolkit for Success” Program	June 2025
Talk at UMD Society for Physics Student Meeting	November 2024
Panel Member and Talk at CU2MIP	April 2024
Talk at UMD PHYS 170 “Professional Physics” Seminar	September 2023
Laboratory Tour for UMD Departmental “Toolkit for Success” Program	July 2023
Outreach at Maryland Day	April 2023
Talk at UMD Graduate Survey Seminar	March 2023
Talk, MSU Science Festival	April 2021
Talk, Making Space for All	June 2020
Talk, Astronomy on Tap Lansing	October 2019, August 2021
Coordinator for ASPIRE Workshop for High School Women, OSU	July 2015-June 2019
Volunteer Judge, Ohio State Science Day	2015-2019
Talk, Columbus Science Pub	May 2018
Talk, The Wellington School, Columbus, OH	April 2018
Volunteer Judge, OSU Denman Undergraduate Research Forum	2016

FULL PUBLICATION LIST

203. R. Abbasi et al. “Evidence for neutrino emission from X-ray Bright Seyfert Galaxies in the Southern Hemisphere using Enhanced Starting Track Events with IceCube”. In: (Feb. 2026). arXiv: 2602.10208 [astro-ph.HE].
202. R. Abbasi et al. “Deep Search for Joint Sources of Gravitational Waves and High-Energy Neutrinos with IceCube During the Third Observing Run of LIGO and Virgo”. In: (Jan. 2026). arXiv: 2601.07595 [astro-ph.HE].
201. R. Abbasi et al. “Constraining the Prompt Atmospheric Neutrino Flux Combining IceCube’s Cascade and Track Samples”. In: (Dec. 2025). arXiv: 2512.17760 [astro-ph.HE].
200. J. Abhir et al. “Prompt Searches for Very-high-energy γ -Ray Counterparts to IceCube Astrophysical Neutrino Alerts”. In: *Astrophys. J.* 997.2 (2026), p. 141. DOI: 10.3847/1538-4357/ae2c4e. arXiv: 2512.16562 [astro-ph.HE].
199. R. Abbasi et al. “Limits on GeV-scale WIMP Annihilation in Dwarf Spheroidals with IceCube DeepCore”. In: (Nov. 2025). arXiv: 2511.19385 [astro-ph.HE].
198. R. Abbasi et al. “Search for GeV-scale Dark Matter from the Galactic Center with IceCube-DeepCore”. In: (Nov. 2025). arXiv: 2511.00918 [astro-ph.HE].

197. N. Alden et al. “Observation of In-ice Askaryan Radiation from High-Energy Cosmic Rays”. In: (Oct. 2025). arXiv: 2510.21104 [astro-ph.HE].
196. R. Abbasi et al. “Evidence for Neutrino Emission from X-ray Bright Active Galactic Nuclei with IceCube”. In: (Oct. 2025). arXiv: 2510.13403 [astro-ph.HE].
195. R. Abbasi et al. “Limiting the Parameter Space for Unstable eV-scale Neutrinos Using IceCube Data”. In: (Sept. 2025). arXiv: 2510.00209 [hep-ex].
194. M. F. H. Seikh et al. “Techniques for Continuous Wave Identification and Filtering in the Askaryan Radio Array”. In: *PoS ICRC2025* (2025), p. 1171. DOI: 10.22323/1.501.1171. arXiv: 2509.20735 [astro-ph.IM].
193. M. Muzio et al. “First Array-Wide Search for Diffuse UHE Neutrinos with the Askaryan Radio Array”. In: *PoS ICRC2025* (2025), p. 1126. DOI: 10.22323/1.501.1126. arXiv: 2509.16414 [astro-ph.HE].
192. A. Bishop et al. “High-Fidelity Simulations of the Full Askaryan Radio Array and its Sensitivity to Ultra-High Energy Neutrinos”. In: *PoS ICRC2025* (2025), p. 991. DOI: 10.22323/1.501.0991. arXiv: 2509.16425 [astro-ph.HE].
191. N. Alden et al. “Study of an Isolated Double-pulse Cosmic Ray Candidate Recorded with the Askaryan Radio Array”. In: *PoS ICRC2025* (2025), p. 977. DOI: 10.22323/1.501.0977. arXiv: 2509.14407 [astro-ph.HE].
190. R. Abbasi et al. “Physics potential of the IceCube Upgrade for atmospheric neutrino oscillations”. In: (Sept. 2025). arXiv: 2509.13066 [hep-ex].
189. R. Abbasi et al. “Identification and Denoising of Radio Signals from Cosmic-Ray Air Showers using Convolutional Neural Networks”. In: (Aug. 2025). arXiv: 2508.14711 [hep-ex].
188. N. Alden et al. “First Ultra High Energy Neutrino Search with a Hybrid Phased and Traditional Detector in the Askaryan Radio Array”. In: (Aug. 2025). arXiv: 2508.05973 [astro-ph.HE].
187. R. Abbasi et al. “The LED calibration systems for the mDOM and D-Egg sensor modules of the IceCube Upgrade: Design, production, testing and use in module calibration”. In: *JINST* 20.11 (2025), P11026. DOI: 10.1088/1748-0221/20/11/P11026. arXiv: 2508.03822 [astro-ph.IM].
186. R. Abbasi et al. “Improved measurements of the TeV–PeV extragalactic neutrino spectrum from joint analyses of IceCube tracks and cascades”. In: (July 2025). arXiv: 2507.22234 [astro-ph.HE].
185. R. Abbasi et al. “Evidence for a Spectral Break or Curvature in the Spectrum of Astrophysical Neutrinos from 5 TeV–10 PeV”. In: (July 2025). arXiv: 2507.22233 [astro-ph.HE].
184. R. Abbasi et al. “Search for High-Energy Neutrinos From the Sun Using Ten Years of IceCube Data”. In: (July 2025). arXiv: 2507.08457 [hep-ex].
183. R. Abbasi et al. “The IceCube Collaboration – Contributions to the 39th International Cosmic Ray Conference (ICRC2025)”. In: *39th International Cosmic Ray Conference*. July 2025. arXiv: 2507.08666 [astro-ph.HE].
182. R. Abbasi et al. “A search for extremely-high-energy neutrinos with IceCube and implications for the ultra-high-energy cosmic-ray proton fraction”. In: *PoS ICRC2025* (2025), p. 1122. DOI: 10.22323/1.501.1122. arXiv: 2507.07497 [astro-ph.HE].
181. P. Koundal et al. “Machine learning driven reconstruction of cosmic-ray air showers for next generation radio arrays”. In: *PoS ICRC2025* (2025), p. 309. DOI: 10.22323/1.501.0309. arXiv: 2507.07219 [astro-ph.HE].

180. R. Abbasi et al. “All-sky Neutrino Point-source Search with IceCube Combined Track and Cascade Data”. In: *Astrophys. J.* 995.1 (2025), p. 11. DOI: 10.3847/1538-4357/ae113f. arXiv: 2507.07275 [astro-ph.HE].
179. P. Windischhofer et al. “Observation of Broadband In-ice Radiation from Impacting High-Energy Cosmic Rays”. In: *PoS ICRC2025* (2025), p. 1213. DOI: 10.22323/1.501.1213. arXiv: 2507.06151 [astro-ph.HE].
178. R. Abbasi et al. “A Search for Millimeter-bright Blazars as Astrophysical Neutrino Sources”. In: *Astrophys. J.* 999.1 (2026), p. 98. DOI: 10.3847/1538-4357/ae31e2. arXiv: 2507.03989 [astro-ph.HE].
177. R. Abbasi et al. “Measurement of the mean number of muons with energies above 500 GeV in air showers detected with the IceCube Neutrino Observatory”. In: *Phys. Rev. D* 112.8 (2025), p. 082004. DOI: 10.1103/lrjy-3hht. arXiv: 2506.19241 [hep-ex].
176. S. Agarwal et al. “Radio emission from airplanes as observed with RNO-G”. In: *JINST* 20.11 (2025), P11015. DOI: 10.1088/1748-0221/20/11/P11015. arXiv: 2506.17522 [hep-ex].
175. R. Abbasi et al. “GollumFit: An IceCube Open-Source Framework for Binned-Likelihood Neutrino Telescope Analyses”. In: (June 2025). arXiv: 2506.04491 [hep-ex].
174. R. Abbasi et al. “Fast low energy reconstruction using Convolutional Neural Networks”. In: *JINST* 21.02 (2026), P02020. DOI: 10.1088/1748-0221/21/02/P02020. arXiv: 2505.16777 [astro-ph.HE].
173. R. Abbasi et al. “Probing the PeV region in the astrophysical neutrino spectrum using $\nu\mu$ from the Southern sky”. In: *Phys. Rev. D* 112.1 (2025), p. 012022. DOI: 10.1103/2hnq-1fsx. arXiv: 2502.19776 [astro-ph.HE].
172. R. Abbasi et al. “Seasonal variations of the atmospheric muon neutrino spectrum measured with IceCube”. In: *Eur. Phys. J. C* 85.12 (2025), p. 1368. DOI: 10.1140/epjc/s10052-025-14844-0. arXiv: 2502.17890 [astro-ph.HE].
171. R. Abbasi et al. “Search for Heavy Neutral Leptons with IceCube DeepCore”. In: (Feb. 2025). arXiv: 2502.09454 [hep-ex].
170. A. Acharyya et al. “VERITAS and Multiwavelength Observations of the Blazar B3 2247+381 in Response to an IceCube Neutrino Alert”. In: *Astrophys. J.* 982.2 (2025), p. 80. DOI: 10.3847/1538-4357/adb30c. arXiv: 2502.03853 [astro-ph.HE].
169. R. Abbasi et al. “Search for Extremely-High-Energy Neutrinos and First Constraints on the Ultrahigh-Energy Cosmic-Ray Proton Fraction with IceCube”. In: *Phys. Rev. Lett.* 135.3 (2025), p. 031001. DOI: 10.1103/PhysRevLett.135.031001. arXiv: 2502.01963 [astro-ph.HE].
168. R. Abbasi et al. “Time-integrated Southern-sky Neutrino Source Searches with 10 yr of IceCube Starting-track Events at Energies Down to 1 TeV”. In: *Astrophys. J.* 998.1 (2026), p. 37. DOI: 10.3847/1538-4357/ae2c86. arXiv: 2501.16440 [astro-ph.HE].
167. R. Abbasi et al. “Search for Neutrino Doublets and Triplets Using 11.4 yr of IceCube Data”. In: *Astrophys. J.* 981.2 (2025), p. 159. DOI: 10.3847/1538-4357/adb312. arXiv: 2501.09276 [astro-ph.HE].
166. R. Abbasi et al. “Search for dark matter from the center of the Earth with 10 years of IceCube data”. In: *Eur. Phys. J. C* 85.5 (2025), p. 490. DOI: 10.1140/epjc/s10052-025-14144-7. arXiv: 2412.12972 [astro-ph.HE].

165. R. Abbasi et al. “Observation of Cosmic-Ray Anisotropy in the Southern Hemisphere with 12 yr of Data Collected by the IceCube Neutrino Observatory”. In: *Astrophys. J.* 981.2 (2025), p. 182. DOI: 10.3847/1538-4357/adb1de. arXiv: 2412.05046 [astro-ph.HE].
164. S. Ali et al. “Askaryan Radio Array: searching for the highest energy neutrinos”. In: *Eur. Phys. J. ST* 234.16 (2025), pp. 5065–5075. DOI: 10.1140/epjs/s11734-025-01708-7. arXiv: 2411.01761 [astro-ph.HE].
163. S. Ali et al. “A search for the ultra high energy neutrinos with the low threshold phased array trigger system of the Askaryan Radio Array”. In: (Sept. 2024). arXiv: 2409.19847 [astro-ph.HE].
162. S. Ali et al. “ARA-Next: a new DAQ and trigger architecture for the Askaryan Radio Array”. In: (Sept. 2024). arXiv: 2409.07634 [astro-ph.HE].
161. S. Ali et al. “Progress towards an array-wide diffuse UHE neutrino search with the Askaryan Radio Array”. In: (Sept. 2024). arXiv: 2409.03854 [astro-ph.IM].
160. R. Abbasi et al. “Search for a light sterile neutrino with 7.5 years of IceCube DeepCore data”. In: *Phys. Rev. D* 110.7 (2024), p. 072007. DOI: 10.1103/PhysRevD.110.072007. arXiv: 2407.01314 [hep-ex].
159. R. Abbasi et al. “Probing the Connection between IceCube Neutrinos and MOJAVE AGN”. In: *Astrophys. J.* 973.2 (2024), p. 97. DOI: 10.3847/1538-4357/ad643d. arXiv: 2407.01351 [astro-ph.HE].
158. R. Abbasi et al. “IceCube Search for Neutrino Emission from X-Ray Bright Seyfert Galaxies”. In: *Astrophys. J.* 988.1 (2025), p. 141. DOI: 10.3847/1538-4357/addd05. arXiv: 2406.07601 [astro-ph.HE].
157. R. Abbasi et al. “Search for Neutrino Emission from Hard X-Ray AGN with IceCube”. In: *Astrophys. J.* 981.2 (2025), p. 131. DOI: 10.3847/1538-4357/ada94b. arXiv: 2406.06684 [astro-ph.HE].
156. R. Abbasi et al. “Exploration of mass splitting and muon/tau mixing parameters for an eV-scale sterile neutrino with IceCube”. In: *Phys. Lett. B* 858 (2024), p. 139077. DOI: 10.1016/j.physletb.2024.139077. arXiv: 2406.00905 [hep-ex].
155. S. Ali et al. “Modeling the refractive index profile $n(z)$ of polar ice for ultra-high energy neutrino experiments”. In: (June 2024). arXiv: 2406.00857 [astro-ph.IM].
154. R. Abbasi et al. “Search for an eV-Scale Sterile Neutrino Using Improved High-Energy $\nu\mu$ Event Reconstruction in IceCube”. In: *Phys. Rev. Lett.* 133.20 (2024), p. 201804. DOI: 10.1103/PhysRevLett.133.201804. arXiv: 2405.08070 [hep-ex].
153. R. Abbasi et al. “Methods and stability tests associated with the sterile neutrino search using improved high-energy $\nu\mu$ event reconstruction in IceCube”. In: *Phys. Rev. D* 110.9 (2024), p. 092009. DOI: 10.1103/PhysRevD.110.092009. arXiv: 2405.08077 [hep-ex].
152. R. Alfaro et al. “Search for Joint Multimessenger Signals from Potential Galactic Cosmic-Ray Accelerators with HAWC and IceCube”. In: *Astrophys. J.* 976.1 (2024), p. 8. DOI: 10.3847/1538-4357/ad812f. arXiv: 2405.03817 [astro-ph.HE].
151. R. Abbasi et al. “Measurement of Atmospheric Neutrino Oscillation Parameters Using Convolutional Neural Networks with 9.3 Years of Data in IceCube DeepCore”. In: *Phys. Rev. Lett.* 134.9 (2025), p. 091801. DOI: 10.1103/PhysRevLett.134.091801. arXiv: 2405.02163 [hep-ex].
150. R. Abbasi et al. “Acceptance Tests of more than 10 000 Photomultiplier Tubes for the multi-PMT Digital Optical Modules of the IceCube Upgrade”. In: *JINST* 19.07 (2024), P07038. DOI: 10.1088/1748-0221/19/07/P07038. arXiv: 2404.19589 [astro-ph.IM].

149. S. Agarwal et al. “Solar flare observations with the Radio Neutrino Observatory Greenland (RNO-G)”. In: *Astropart. Phys.* 164 (2025), p. 103024. DOI: 10.1016/j.astropartphys.2024.103024. arXiv: 2404.14995 [astro-ph.SR].
148. R. Abbasi et al. “Improved modeling of in-ice particle showers for IceCube event reconstruction”. In: *JINST* 19.06 (2024), P06026. DOI: 10.1088/1748-0221/19/06/P06026. arXiv: 2403.02470 [astro-ph.HE].
147. R. Abbasi et al. “Characterization of the astrophysical diffuse neutrino flux using starting track events in IceCube”. In: *Phys. Rev. D* 110.2 (2024), p. 022001. DOI: 10.1103/PhysRevD.110.022001. arXiv: 2402.18026 [astro-ph.HE].
146. R. Abbasi et al. “Citizen science for IceCube: Name that Neutrino”. In: *Eur. Phys. J. Plus* 139.6 (2024), p. 533. DOI: 10.1140/epjp/s13360-024-05179-y. arXiv: 2401.11994 [astro-ph.HE].
145. R. Abbasi et al. “Search for 10–1000 GeV Neutrinos from Gamma-Ray Bursts with IceCube”. In: *Astrophys. J.* 964.2 (2024), p. 126. DOI: 10.3847/1538-4357/ad220b. arXiv: 2312.11515 [astro-ph.HE].
144. R. Abbasi et al. “All-sky Search for Transient Astrophysical Neutrino Emission with 10 Years of IceCube Cascade Events”. In: *Astrophys. J.* 967.1 (2024), p. 48. DOI: 10.3847/1538-4357/ad3730. arXiv: 2312.05362 [astro-ph.HE].
143. N. Kurahashi Neilson et al. “Highlights from the IceCube Neutrino Observatory”. In: *PoS ICRC2023* (2024), p. 017. DOI: 10.22323/1.444.0017. arXiv: 2310.12840 [astro-ph.HE].
142. R. Abbasi et al. “Search for Continuous and Transient Neutrino Emission Associated with IceCube’s Highest-energy Tracks: An 11 yr Analysis”. In: *Astrophys. J.* 964.1 (2024), p. 40. DOI: 10.3847/1538-4357/ad18d6. arXiv: 2309.12130 [astro-ph.HE].
141. M. F. H. Seikh et al. “Calibration and Physics with ARA Station 1: A Unique Askaryan Radio Array Detector”. In: *PoS ICRC2023* (2023), p. 1163. DOI: 10.22323/1.444.1163. arXiv: 2308.07292 [astro-ph.IM].
140. R. Abbasi et al. “Search for Galactic Core-collapse Supernovae in a Decade of Data Taken with the IceCube Neutrino Observatory”. In: *Astrophys. J.* 961.1 (2024), p. 84. DOI: 10.3847/1538-4357/ad07d1. arXiv: 2308.01172 [astro-ph.HE].
139. W. Hou et al. “Sensitivity of the IceCube-Gen2 Surface Array for Cosmic-Ray Anisotropy Studies”. In: *PoS ICRC2023* (2023), p. 354. DOI: 10.22323/1.444.0354. arXiv: 2307.14655 [astro-ph.HE].
138. R. Abbasi et al. “IceCube search for neutrinos from novae”. In: *PoS ICRC2023* (2023), p. 1560. DOI: 10.22323/1.444.1560. arXiv: 2307.15372 [astro-ph.HE].
137. M. Dittmer et al. “Performance studies on new 4” photomultiplier types intended for IceCube-Gen2 optical modules”. In: *PoS ICRC2023* (2023), p. 985. DOI: 10.22323/1.444.0985. arXiv: 2307.14589 [astro-ph.IM].
136. F. Schlüter et al. “Estimating the coincidence rate between the optical and radio array of IceCube-Gen2”. In: *PoS ICRC2023* (2023), p. 1022. DOI: 10.22323/1.444.1022. arXiv: 2308.00961 [astro-ph.HE].
135. A. Ishihara et al. “The next generation neutrino telescope: IceCube-Gen2”. In: *PoS ICRC2023* (2023), p. 994. DOI: 10.22323/1.444.0994. arXiv: 2308.09427 [astro-ph.HE].
134. Y. Makino et al. “Mechanical design of the optical modules intended for IceCube-Gen2”. In: *PoS ICRC2023* (2023), p. 979. DOI: 10.22323/1.444.0979. arXiv: 2308.09786 [astro-ph.IM].

133. T. Glüsenkamp et al. “VAE-based latent-space classification of RNO-G data”. In: *PoS ICRC2023* (2023), p. 1056. DOI: 10.22323/1.444.1056. arXiv: 2309.16401 [astro-ph.HE].
132. N. Lad et al. “Sensitivity of IceCube-Gen2 to measure flavor composition of Astrophysical neutrinos”. In: *PoS ICRC2023* (2023), p. 1123. DOI: 10.22323/1.444.1123. arXiv: 2308.15220 [astro-ph.HE].
131. R. Abbasi et al. “Searching for High-energy Neutrino Emission from Seyfert Galaxies in the Northern Sky with IceCube”. In: *PoS ICRC2023* (2023), p. 1052. DOI: 10.22323/1.444.1052. arXiv: 2308.00024 [astro-ph.HE].
130. A. Coleman et al. “Enhancing the Sensitivity of RNO-G Using a Machine-learning Based Trigger”. In: *PoS ICRC2023* (2023), p. 1100. DOI: 10.22323/1.444.1100.
129. E. Oberla et al. “Low-Power Radiofrequency Systems for the RNO-G Project”. In: *PoS ICRC2023* (2023), p. 1171. DOI: 10.22323/1.444.1171.
128. M. S. Muzio et al. “Multimessenger Potential of the Radio Neutrino Observatory in Greenland”. In: *PoS ICRC2023* (2023), p. 1485. DOI: 10.22323/1.444.1485. arXiv: 2308.07224 [astro-ph.HE].
127. R. Abbasi et al. “Search for decoherence from quantum gravity with atmospheric neutrinos”. In: *Nature Phys.* 20.6 (2024), pp. 913–920. DOI: 10.1038/s41567-024-02436-w. arXiv: 2308.00105 [hep-ex].
126. R. Abbasi et al. “The IceCube Collaboration – Contributions to the 38th International Cosmic Ray Conference (ICRC2023)”. In: *38th International Cosmic Ray Conference*. July 2023. arXiv: 2307.13047 [astro-ph.HE].
125. R. Abbasi et al. “Search for Extended Sources of Neutrino Emission in the Galactic Plane with IceCube”. In: *Astrophys. J.* 956.1 (2023), p. 20. DOI: 10.3847/1538-4357/acf713. arXiv: 2307.07576 [astro-ph.HE].
124. R. Abbasi et al. “Observation of high-energy neutrinos from the Galactic plane”. In: *Science* 380.6652 (2023), adc9818. DOI: 10.1126/science.adc9818. arXiv: 2307.04427 [astro-ph.HE].
123. J. Henrichs et al. “Searching for air showers with RNO-G”. In: *PoS ARENA2022* (2023), p. 007. DOI: 10.22323/1.424.0007.
122. J. A. Aguilar et al. “The Radio Neutrino Observatory Greenland RNO-G: Status update”. In: *PoS ARENA2022* (2023), p. 005. DOI: 10.22323/1.424.0005.
121. R. Abbasi et al. “Search for Correlations of High-energy Neutrinos Detected in IceCube with Radio-bright AGN and Gamma-Ray Emission from Blazars”. In: *Astrophys. J.* 954.1 (2023), p. 75. DOI: 10.3847/1538-4357/acdfcb. arXiv: 2304.12675 [astro-ph.HE].
120. R. Abbasi et al. “Measurement of atmospheric neutrino mixing with improved IceCube DeepCore calibration and data processing”. In: *Phys. Rev. D* 108.1 (2023), p. 012014. DOI: 10.1103/PhysRevD.108.012014. arXiv: 2304.12236 [hep-ex].
119. R. Abbasi et al. “IceCat-1: The IceCube Event Catalog of Alert Tracks”. In: *Astrophys. J. Suppl.* 269.1 (2023), p. 25. DOI: 10.3847/1538-4365/acfa95. arXiv: 2304.01174 [astro-ph.HE].
118. R. Abbasi et al. “A Search for IceCube Sub-TeV Neutrinos Correlated with Gravitational-wave Events Detected By LIGO/Virgo”. In: *Astrophys. J.* 959.2 (2023). [Erratum: *Astrophys. J.* 971, 192 (2024)], p. 96. DOI: 10.3847/1538-4357/aceefc. arXiv: 2303.15970 [astro-ph.HE].
117. R. Abbasi et al. “Search for neutrino lines from dark matter annihilation and decay with IceCube”. In: *Phys. Rev. D* 108.10 (2023), p. 102004. DOI: 10.1103/PhysRevD.108.102004. arXiv: 2303.13663 [astro-ph.HE].

116. R. Abbasi et al. “Observation of seasonal variations of the flux of high-energy atmospheric neutrinos with IceCube”. In: *Eur. Phys. J. C* 83.9 (2023), p. 777. DOI: 10.1140/epjc/s10052-023-11679-5. arXiv: 2303.04682 [astro-ph.HE].
115. R. Abbasi et al. “Constraining High-energy Neutrino Emission from Supernovae with IceCube”. In: *Astrophys. J. Lett.* 949.1 (2023), p. L12. DOI: 10.3847/2041-8213/acd2c9. arXiv: 2303.03316 [astro-ph.HE].
114. J. A. Aguilar et al. “Triboelectric backgrounds to radio-based polar ultra-high energy neutrino (UHEN) experiments”. In: *Astropart. Phys.* 145 (2023), p. 102790. DOI: 10.1016/j.astropartphys.2022.102790.
113. L. Pyras et al. “The Radio Neutrino Observatory Greenland: Status Update and Prospect for Air Showers”. In: *PoS ECRS* (2023), p. 088. DOI: 10.22323/1.423.0088.
112. R. Abbasi et al. “Limits on Neutrino Emission from GRB 221009A from MeV to PeV Using the IceCube Neutrino Observatory”. In: *Astrophys. J. Lett.* 946.1 (2023). [Erratum: *Astrophys.J.Lett.* 970, L43 (2024), Erratum: *Astrophys.J.* 970, L43 (2024)], p. L26. DOI: 10.3847/2041-8213/acc077. arXiv: 2302.05459 [astro-ph.HE].
111. R. Abbasi et al. “D-Egg: a dual PMT optical module for IceCube”. In: *JINST* 18.04 (2023), P04014. DOI: 10.1088/1748-0221/18/04/P04014. arXiv: 2212.14526 [astro-ph.IM].
110. R. Abbasi et al. “A Search for Coincident Neutrino Emission from Fast Radio Bursts with Seven Years of IceCube Cascade Events”. In: *Astrophys. J.* 946.2 (2023), p. 80. DOI: 10.3847/1538-4357/acbea0. arXiv: 2212.06702 [astro-ph.HE].
109. R. Abbasi et al. “Search for sub-TeV Neutrino Emission from Novae with IceCube-DeepCore”. In: *Astrophys. J.* 953.2 (2023). [Erratum: *Astrophys.J.* 971, (2024)], p. 160. DOI: 10.3847/1538-4357/acdc1b. arXiv: 2212.06810 [astro-ph.HE].
108. R. Abbasi et al. “Searches for Neutrinos from Large High Altitude Air Shower Observatory Ultra-high-energy γ -Ray Sources Using the IceCube Neutrino Observatory”. In: *Astrophys. J. Lett.* 945.1 (2023), p. L8. DOI: 10.3847/2041-8213/acb933. arXiv: 2211.14184 [astro-ph.HE].
107. R. Abbasi et al. “Evidence for neutrino emission from the nearby active galaxy NGC 1068”. In: *Science* 378.6619 (2022), pp. 538–543. DOI: 10.1126/science.abg3395. arXiv: 2211.09972 [astro-ph.HE].
106. R. Abbasi et al. “Constraints on Populations of Neutrino Sources from Searches in the Directions of IceCube Neutrino Alerts”. In: *Astrophys. J.* 951.1 (2023), p. 45. DOI: 10.3847/1538-4357/acd2ca. arXiv: 2210.04930 [astro-ph.HE].
105. R. Abbasi et al. “Graph Neural Networks for low-energy event classification & reconstruction in IceCube”. In: *JINST* 17.11 (2022), P11003. DOI: 10.1088/1748-0221/17/11/P11003. arXiv: 2209.03042 [hep-ex].
104. R. Abbasi et al. “IceCube Search for Neutrinos Coincident with Gravitational Wave Events from LIGO/Virgo Run O3”. In: *Astrophys. J.* 944.1 (2023), p. 80. DOI: 10.3847/1538-4357/aca5fc. arXiv: 2208.09532 [astro-ph.HE].
103. R. Abbasi et al. “Search for Astrophysical Neutrinos from 1FLE Blazars with IceCube”. In: *Astrophys. J.* 938.1 (2022), p. 38. DOI: 10.3847/1538-4357/ac8de4. arXiv: 2207.04946 [astro-ph.HE].
102. R. Abbasi et al. “Searching for High-energy Neutrino Emission from Galaxy Clusters with IceCube”. In: *Astrophys. J. Lett.* 938.2 (2022), p. L11. DOI: 10.3847/2041-8213/ac966b. arXiv: 2206.02054 [astro-ph.HE].

101. R. Abbasi et al. “Searches for connections between dark matter and high-energy neutrinos with IceCube”. In: *JCAP* 10 (2023), p. 003. DOI: 10.1088/1475-7516/2023/10/003. arXiv: 2205.12950 [hep-ex].
100. R. Abbasi et al. “Searches for Neutrinos from Gamma-Ray Bursts Using the IceCube Neutrino Observatory”. In: *Astrophys. J.* 939.2 (2022), p. 116. DOI: 10.3847/1538-4357/ac9785. arXiv: 2205.11410 [astro-ph.HE].
99. R. Abbasi et al. “Framework and tools for the simulation and analysis of the radio emission from air showers at IceCube”. In: *JINST* 17.06 (2022), P06026. DOI: 10.1088/1748-0221/17/06/P06026. arXiv: 2205.02258 [astro-ph.HE].
98. R. Abbasi et al. “Search for Unstable Sterile Neutrinos with the IceCube Neutrino Observatory”. In: *Phys. Rev. Lett.* 129.15 (2022), p. 151801. DOI: 10.1103/PhysRevLett.129.151801. arXiv: 2204.00612 [hep-ex].
97. M. Ackermann et al. “High-energy and ultra-high-energy neutrinos: A Snowmass white paper”. In: *JHEAp* 36 (2022), pp. 55–110. DOI: 10.1016/j.jheap.2022.08.001. arXiv: 2203.08096 [hep-ph].
96. R. Abbasi et al. “Low energy event reconstruction in IceCube DeepCore”. In: *Eur. Phys. J. C* 82.9 (2022), p. 807. DOI: 10.1140/epjc/s10052-022-10721-2. arXiv: 2203.02303 [hep-ex].
95. R. Abbasi et al. “Search for High-energy Neutrino Emission from Galactic X-Ray Binaries with IceCube”. In: *Astrophys. J. Lett.* 930.2 (2022), p. L24. DOI: 10.3847/2041-8213/ac67d8. arXiv: 2202.11722 [astro-ph.HE].
94. J. van Santen, B. A. Clark, R. Halliday, S. Hallmann, and A. Nelles. “toise: a framework to describe the performance of high-energy neutrino detectors”. In: *JINST* 17.08 (2022), T08009. DOI: 10.1088/1748-0221/17/08/T08009. arXiv: 2202.11120 [astro-ph.IM].
93. P. Allison et al. “Low-threshold ultrahigh-energy neutrino search with the Askaryan Radio Array”. In: *Phys. Rev. D* 105.12 (2022), p. 122006. DOI: 10.1103/PhysRevD.105.122006. arXiv: 2202.07080 [astro-ph.HE].
92. R. Abbasi et al. “Density of GeV muons in air showers measured with IceTop”. In: *Phys. Rev. D* 106.3 (2022), p. 032010. DOI: 10.1103/PhysRevD.106.032010. arXiv: 2201.12635 [hep-ex].
91. J. A. Aguilar et al. “In situ, broadband measurement of the radio frequency attenuation length at Summit Station, Greenland”. In: *J. Glaciol.* 68.272 (2022), pp. 1234–1242. DOI: 10.1017/jog.2022.40. arXiv: 2201.07846 [astro-ph.IM].
90. A. Albert et al. “Search for Spatial Correlations of Neutrinos with Ultra-high-energy Cosmic Rays”. In: *Astrophys. J.* 934.2 (2022), p. 164. DOI: 10.3847/1538-4357/ac6def. arXiv: 2201.07313 [astro-ph.HE].
89. R. Abbasi et al. “Strong Constraints on Neutrino Nonstandard Interactions from TeV-Scale ν_u Disappearance at IceCube”. In: *Phys. Rev. Lett.* 129.1 (2022), p. 011804. DOI: 10.1103/PhysRevLett.129.011804. arXiv: 2201.03566 [hep-ex].
88. R. Abbasi et al. “Improved Characterization of the Astrophysical Muon–neutrino Flux with 9.5 Years of IceCube Data”. In: *Astrophys. J.* 928.1 (2022), p. 50. DOI: 10.3847/1538-4357/ac4d29. arXiv: 2111.10299 [astro-ph.HE].
87. R. Abbasi et al. “Search for neutrino emission from cores of active galactic nuclei”. In: *Phys. Rev. D* 106.2 (2022), p. 022005. DOI: 10.1103/PhysRevD.106.022005. arXiv: 2111.10169 [astro-ph.HE].

86. R. Abbasi et al. “Search for GeV-scale dark matter annihilation in the Sun with IceCube DeepCore”. In: *Phys. Rev. D* 105.6 (2022), p. 062004. DOI: 10.1103/PhysRevD.105.062004. arXiv: 2111.09970 [astro-ph.HE].
85. R. Abbasi et al. “Search for quantum gravity using astrophysical neutrino flavour with IceCube”. In: *Nature Phys.* 18.11 (2022), pp. 1287–1292. DOI: 10.1038/s41567-022-01762-1. arXiv: 2111.04654 [hep-ex].
84. R. Abbasi et al. “Search for Relativistic Magnetic Monopoles with Eight Years of IceCube Data”. In: *Phys. Rev. Lett.* 128.5 (2022), p. 051101. DOI: 10.1103/PhysRevLett.128.051101. arXiv: 2109.13719 [astro-ph.HE].
83. R. Abbasi et al. “Search for Multi-flare Neutrino Emissions in 10 yr of IceCube Data from a Catalog of Sources”. In: *Astrophys. J. Lett.* 920.2 (2021), p. L45. DOI: 10.3847/2041-8213/ac2c7b. arXiv: 2109.05818 [astro-ph.HE].
82. V. A. Acciari et al. “Searching for VHE gamma-ray emission associated with IceCube neutrino alerts using FACT, H.E.S.S., MAGIC, and VERITAS”. In: *PoS ICRC2021* (2021), p. 960. DOI: 10.22323/1.395.0960. arXiv: 2109.04350 [astro-ph.HE].
81. B. Clark. “The IceCube-Gen2 Neutrino Observatory”. In: *JINST* 16.10 (2021), p. C10007. DOI: 10.1088/1748-0221/16/10/C10007. arXiv: 2108.05292 [astro-ph.HE].
80. H. Ayala et al. “Multimessenger NuEM Alerts with AMON”. In: *PoS ICRC2021* (2021), p. 958. DOI: 10.22323/1.395.0958. arXiv: 2108.04920 [astro-ph.HE].
79. R. Abbasi et al. “Simulation study for the future IceCube-Gen2 surface array”. In: *PoS ICRC2021* (2021), p. 411. DOI: 10.22323/1.395.0411. arXiv: 2108.04307 [astro-ph.HE].
78. R. Abbasi et al. “Searching for neutrino transients below 1 TeV with IceCube”. In: *PoS ICRC2021* (2021), p. 1131. DOI: 10.22323/1.395.1131. arXiv: 2108.01530 [astro-ph.HE].
77. F. G. Schroeder et al. “The Surface Array planned for IceCube-Gen2”. In: *PoS ICRC2021* (2021), p. 407. DOI: 10.22323/1.395.0407. arXiv: 2108.00364 [astro-ph.HE].
76. B. Clark et al. “Characterization of the PeV astrophysical neutrino energy spectrum with IceCube using down-going tracks”. In: *PoS ICRC2021* (2021), p. 1137. DOI: 10.22323/1.395.1137. arXiv: 2107.14298 [astro-ph.HE].
75. R. Abbasi et al. “Searching for time-dependent high-energy neutrino emission from X-ray binaries with IceCube”. In: *PoS ICRC2021* (2021), p. 1136. DOI: 10.22323/1.395.1136. arXiv: 2107.12383 [astro-ph.HE].
74. R. Abbasi et al. “Gravitational Wave Follow-Up Using Low Energy Neutrinos in IceCube DeepCore”. In: *PoS ICRC2021* (2021), p. 939. DOI: 10.22323/1.395.0939. arXiv: 2107.11285 [astro-ph.HE].
73. R. Abbasi et al. “Discrimination of Muons for Mass Composition Studies of Inclined Air Showers Detected with IceTop”. In: *PoS ICRC2021* (2021), p. 212. DOI: 10.22323/1.395.0212. arXiv: 2107.11293 [astro-ph.HE].
72. R. Abbasi et al. “Reconstructing Neutrino Energy using CNNs for GeV Scale IceCube Events”. In: *PoS ICRC2021* (2021), p. 1053. DOI: 10.22323/1.395.1053. arXiv: 2107.11446 [astro-ph.HE].
71. E. Bechtol et al. “Towards Equitable, Diverse, and Inclusive science collaborations: The Multimessenger Diversity Network”. In: *PoS ICRC2021* (2021), p. 1383. DOI: 10.22323/1.395.1383. arXiv: 2107.12179 [physics.ed-ph].
70. R. Abbasi et al. “New flux limit in the low relativistic regime for magnetic monopoles at IceCube”. In: *PoS ICRC2021* (2021), p. 534. DOI: 10.22323/1.395.0534. arXiv: 2107.10548 [astro-ph.HE].

69. L. Halve et al. “Design of an Efficient, High-Throughput Photomultiplier Tube Testing Facility for the IceCube Upgrade”. In: *PoS ICRC2021* (2021), p. 1056. DOI: 10.22323/1.395.1056. arXiv: 2107.09954 [astro-ph.HE].
68. J. Necker et al. “Searching for High-Energy Neutrinos from Core-Collapse Supernovae with IceCube”. In: *PoS ICRC2021* (2021), p. 1116. DOI: 10.22323/1.395.1116. arXiv: 2107.09317 [astro-ph.HE].
67. S. Verpoest et al. “Testing Hadronic Interaction Models with Cosmic Ray Measurements at the IceCube Neutrino Observatory”. In: *PoS ICRC2021* (2021), p. 357. DOI: 10.22323/1.395.0357. arXiv: 2107.09387 [astro-ph.HE].
66. A. Pizzuto et al. “Realtime follow-up of astrophysical transients with the IceCube Neutrino Observatory”. In: *PoS ICRC2021* (2021), p. 952. DOI: 10.22323/1.395.0952. arXiv: 2107.09551 [astro-ph.HE].
65. R. Abbasi et al. “Study of mass composition of cosmic rays with IceTop and IceCube”. In: *PoS ICRC2021* (2021), p. 323. DOI: 10.22323/1.395.0323. arXiv: 2107.09626 [astro-ph.HE].
64. D. Veske et al. “Multi-messenger searches via IceCube’s high-energy neutrinos and gravitational-wave detections of LIGO/Virgo”. In: *PoS ICRC2021* (2021), p. 950. DOI: 10.22323/1.395.0950. arXiv: 2107.09663 [astro-ph.HE].
63. R. Abbasi et al. “Measuring total neutrino cross section with IceCube at intermediate energies (~ 100 GeV to a few TeV)”. In: *PoS ICRC2021* (2021), p. 1132. DOI: 10.22323/1.395.1132. arXiv: 2107.09764 [astro-ph.HE].
62. R. Abbasi et al. “First air-shower measurements with the prototype station of the IceCube surface enhancement”. In: *PoS ICRC2021* (2021), p. 314. DOI: 10.22323/1.395.0314. arXiv: 2107.08750 [astro-ph.HE].
61. R. Abbasi et al. “Sensitivity studies for the IceCube-Gen2 radio array”. In: *PoS ICRC2021* (2021), p. 1183. DOI: 10.22323/1.395.1183. arXiv: 2107.08910 [astro-ph.HE].
60. R. Abbasi et al. “Searches for Neutrinos from Precursors and Afterglows of Gamma-Ray Bursts using the IceCube Neutrino Observatory”. In: *PoS ICRC2021* (2021), p. 1118. DOI: 10.22323/1.395.1118. arXiv: 2107.08870 [astro-ph.HE].
59. R. Abbasi et al. “IceCube Search for High-Energy Neutrinos from Ultra-Luminous Infrared Galaxies”. In: *PoS ICRC2021* (2021), p. 1115. DOI: 10.22323/1.395.1115. arXiv: 2107.08422 [astro-ph.HE].
58. R. Abbasi et al. “Search for high-energy neutrino emission from hard X-ray AGN with IceCube”. In: *JINST* 16.09 (2021), p. C09013. DOI: 10.1088/1748-0221/16/09/C09013. arXiv: 2107.08366 [astro-ph.HE].
57. R. Abbasi et al. “Simulation and sensitivities for a phased IceCube-Gen2 deployment”. In: *PoS ICRC2021* (2021), p. 1186. DOI: 10.22323/1.395.1186. arXiv: 2107.08500 [astro-ph.HE].
56. A. Omeliukh et al. “Optimization of the optical array geometry for IceCube-Gen2”. In: *PoS ICRC2021* (2021), p. 1184. DOI: 10.22323/1.395.1184. arXiv: 2107.08527 [astro-ph.HE].
55. R. Abbasi et al. “Analysis framework for multi-messenger astronomy with IceCube”. In: *PoS ICRC2021* (2021), p. 1098. DOI: 10.22323/1.395.1098. arXiv: 2107.08254 [astro-ph.IM].
54. M. J. Larson et al. “Testing the AGN Radio and Neutrino correlation using the MOJAVE catalog and 10 years of IceCube Data”. In: *PoS ICRC2021* (2021), p. 949. DOI: 10.22323/1.395.0949. arXiv: 2107.08115 [astro-ph.HE].
53. V. Basu et al. “A next-generation optical sensor for IceCube-Gen2”. In: *PoS ICRC2021* (2021), p. 1062. DOI: 10.22323/1.395.1062. arXiv: 2107.08837 [astro-ph.IM].

52. R. Abbasi et al. “The IceCube-Gen2 Collaboration – Contributions to the 37th International Cosmic Ray Conference (ICRC2021)”. In: (July 2021). arXiv: 2107.06968 [astro-ph.HE].
51. R. Abbasi et al. “The IceCube Collaboration – Contributions to the 37th International Cosmic Ray Conference (ICRC2021)”. In: (July 2021). arXiv: 2107.06966 [astro-ph.HE].
50. R. Abbasi et al. “Search for High-energy Neutrinos from Ultraluminous Infrared Galaxies with IceCube”. In: *Astrophys. J.* 926.1 (2022), p. 59. DOI: 10.3847/1538-4357/ac3cb6. arXiv: 2107.03149 [astro-ph.HE].
49. J. A. Aguilar et al. “Reconstructing the neutrino energy for in-ice radio detectors: A study for the Radio Neutrino Observatory Greenland (RNO-G)”. In: *Eur. Phys. J. C* 82.2 (2022), p. 147. DOI: 10.1140/epjc/s10052-022-10034-4. arXiv: 2107.02604 [astro-ph.HE].
48. R. Abbasi et al. “All-flavor constraints on nonstandard neutrino interactions and generalized matter potential with three years of IceCube DeepCore data”. In: *Phys. Rev. D* 104.7 (2021), p. 072006. DOI: 10.1103/PhysRevD.104.072006. arXiv: 2106.07755 [hep-ex].
47. R. Abbasi et al. “Probing neutrino emission at GeV energies from compact binary mergers with the IceCube Neutrino Observatory”. In: *Phys. Rev. D* 113.4 (2026), p. 042003. DOI: 10.1103/bkxc-4116. arXiv: 2105.13160 [astro-ph.HE].
46. M. G. Aartsen et al. “Detection of a particle shower at the Glashow resonance with IceCube”. In: *Nature* 591.7849 (2021). [Erratum: *Nature* 592, E11 (2021)], pp. 220–224. DOI: 10.1038/s41586-021-03256-1. arXiv: 2110.15051 [hep-ex].
45. R. Abbasi et al. “A muon-track reconstruction exploiting stochastic losses for large-scale Cherenkov detectors”. In: *JINST* 16.08 (2021), P08034. DOI: 10.1088/1748-0221/16/08/P08034. arXiv: 2103.16931 [hep-ex].
44. J. A. Aguilar et al. “Triboelectric Backgrounds to radio-based UHE Neutrino Experiments”. In: (Mar. 2021). arXiv: 2103.06079 [astro-ph.IM].
43. R. Abbasi et al. “A Convolutional Neural Network based Cascade Reconstruction for the IceCube Neutrino Observatory”. In: *JINST* 16 (2021), P07041. DOI: 10.1088/1748-0221/16/07/P07041. arXiv: 2101.11589 [hep-ex].
42. R. Abbasi et al. “IceCube Data for Neutrino Point-Source Searches Years 2008-2018”. In: (Jan. 2021). DOI: 10.21234/CPKQ-K003. arXiv: 2101.09836 [astro-ph.HE].
41. R. Abbasi et al. “Search for GeV neutrino emission during intense gamma-ray solar flares with the IceCube Neutrino Observatory”. In: *Phys. Rev. D* 103.10 (2021), p. 102001. DOI: 10.1103/PhysRevD.103.102001. arXiv: 2101.00610 [astro-ph.HE].
40. K. Hughes et al. “Implementing a Low-Threshold Analysis with the Askaryan Radio Array (ARA)”. In: *PoS ICRC2021* (2021), p. 1153. DOI: 10.22323/1.395.1153.
39. M. Kim et al. “A Template-based UHE Neutrino Search Strategy for the Askaryan Radio Array (ARA)”. In: *PoS ICRC2021* (2021), p. 1147. DOI: 10.22323/1.395.1147.
38. P. Dasgupta et al. “The Calibration of the Geometry and Antenna Delay in Askaryan Radio Array Station 4 and 5”. In: *PoS ICRC2021* (2021), p. 1086. DOI: 10.22323/1.395.1086.
37. J. A. Aguilar et al. “Hardware Development for the Radio Neutrino Observatory in Greenland (RNO-G)”. In: *PoS ICRC2021* (2021), p. 1058. DOI: 10.22323/1.395.1058.
36. R. Abbasi et al. “Search for dark matter from the center of the Earth with 8 years of IceCube data”. In: *PoS ICRC2021* (2021), p. 526. DOI: 10.22323/1.395.0526.

35. J. A. Aguilar et al. “Energy reconstruction with the Radio Neutrino Observatory Greenland (RNO-G)”. In: *PoS ICRC2021* (2021), p. 1033. DOI: 10.22323/1.395.1033.
34. Y. Pan et al. “A neural network based UHE neutrino reconstruction method for the Askaryan Radio Array (ARA)”. In: *PoS ICRC2021* (2021), p. 1157. DOI: 10.22323/1.395.1157.
33. J. A. Aguilar et al. “Direction Reconstruction for the Radio Neutrino Observatory Greenland (RNO-G)”. In: *PoS ICRC2021* (2021), p. 1026. DOI: 10.22323/1.395.1026.
32. J. A. Aguilar et al. “The Radio Neutrino Observatory Greenland (RNO-G)”. In: *PoS ICRC2021* (2021), p. 001. DOI: 10.22323/1.395.0001.
31. R. Abbasi et al. “LeptonInjector and LeptonWeighter: A neutrino event generator and weighter for neutrino observatories”. In: *Comput. Phys. Commun.* 266 (2021), p. 108018. DOI: 10.1016/j.cpc.2021.108018. arXiv: 2012.10449 [physics.comp-ph].
30. R. Abbasi et al. “Follow-up of Astrophysical Transients in Real Time with the IceCube Neutrino Observatory”. In: *Astrophys. J.* 910.1 (2021), p. 4. DOI: 10.3847/1538-4357/abe123. arXiv: 2012.04577 [astro-ph.HE].
29. R. Abbasi et al. “A Search for Time-dependent Astrophysical Neutrino Emission with IceCube Data from 2012 to 2017”. In: *Astrophys. J.* 911.1 (2021), p. 67. DOI: 10.3847/1538-4357/abe7e6. arXiv: 2012.01079 [astro-ph.HE].
28. R. Abbasi et al. “First all-flavor search for transient neutrino emission using 3-years of IceCube DeepCore data”. In: *JCAP* 01 (2022), p. 027. DOI: 10.1088/1475-7516/2022/01/027. arXiv: 2011.05096 [astro-ph.HE].
27. R. Abbasi et al. “The IceCube high-energy starting event sample: Description and flux characterization with 7.5 years of data”. In: *Phys. Rev. D* 104 (2021), p. 022002. DOI: 10.1103/PhysRevD.104.022002. arXiv: 2011.03545 [astro-ph.HE].
26. R. Abbasi et al. “Measurement of the high-energy all-flavor neutrino-nucleon cross section with IceCube”. In: *Phys. Rev. D* 104 (2021), p. 022001. DOI: 10.1103/PhysRevD.104.022001. arXiv: 2011.03560 [hep-ex].
25. R. Abbasi et al. “Detection of astrophysical tau neutrino candidates in IceCube”. In: *Eur. Phys. J. C* 82.11 (2022), p. 1031. DOI: 10.1140/epjc/s10052-022-10795-y. arXiv: 2011.03561 [hep-ex].
24. J. A. Aguilar et al. “Design and Sensitivity of the Radio Neutrino Observatory in Greenland (RNO-G)”. In: *JINST* 16.03 (2021). [Erratum: *JINST* 18, E03001 (2023)], P03025. DOI: 10.1088/1748-0221/16/03/P03025. arXiv: 2010.12279 [astro-ph.IM].
23. H. A. Ayala Solares et al. “Multimessenger Gamma-Ray and Neutrino Coincidence Alerts Using HAWC and IceCube Subthreshold Data”. In: *Astrophys. J.* 906 (2021), p. 63. DOI: 10.3847/1538-4357/abcaa4. arXiv: 2008.10616 [astro-ph.HE].
22. M. G. Aartsen et al. “IceCube-Gen2: the window to the extreme Universe”. In: *J. Phys. G* 48.6 (2021), p. 060501. DOI: 10.1088/1361-6471/abbd48. arXiv: 2008.04323 [astro-ph.HE].
21. M. G. Aartsen et al. “Measurements of the time-dependent cosmic-ray Sun shadow with seven years of IceCube data: Comparison with the Solar cycle and magnetic field models”. In: *Phys. Rev. D* 103.4 (2021), p. 042005. DOI: 10.1103/PhysRevD.103.042005. arXiv: 2006.16298 [astro-ph.HE].
20. M. G. Aartsen et al. “Cosmic ray spectrum from 250 TeV to 10 PeV using IceTop”. In: *Phys. Rev. D* 102 (2020), p. 122001. DOI: 10.1103/PhysRevD.102.122001. arXiv: 2006.05215 [astro-ph.HE].
19. M. G. Aartsen et al. “Searching for eV-scale sterile neutrinos with eight years of atmospheric neutrinos at the IceCube Neutrino Telescope”. In: *Phys. Rev. D* 102.5 (2020), p. 052009. DOI: 10.1103/PhysRevD.102.052009. arXiv: 2005.12943 [hep-ex].

18. M. G. Aartsen et al. “eV-Scale Sterile Neutrino Search Using Eight Years of Atmospheric Muon Neutrino Data from the IceCube Neutrino Observatory”. In: *Phys. Rev. Lett.* 125.14 (2020), p. 141801. DOI: 10.1103/PhysRevLett.125.141801. arXiv: 2005.12942 [hep-ex].
17. J. A. Rolla et al. “Evolving Antennas for Ultra-High Energy Neutrino Detection”. In: *PoS ICRC2019* (2021), p. 992. DOI: 10.22323/1.358.0992. arXiv: 2005.07772 [astro-ph.IM].
16. M. G. Aartsen et al. “IceCube Search for Neutrinos Coincident with Compact Binary Mergers from LIGO-Virgo’s First Gravitational-wave Transient Catalog”. In: *Astrophys. J. Lett.* 898.1 (2020), p. L10. DOI: 10.3847/2041-8213/ab9d24. arXiv: 2004.02910 [astro-ph.HE].
15. M. G. Aartsen et al. “IceCube Search for High-Energy Neutrino Emission from TeV Pulsar Wind Nebulae”. In: *Astrophys. J.* 898.2 (2020), p. 117. DOI: 10.3847/1538-4357/ab9fa0. arXiv: 2003.12071 [astro-ph.HE].
14. A. Albert et al. “Combined search for neutrinos from dark matter self-annihilation in the Galactic Center with ANTARES and IceCube”. In: *Phys. Rev. D* 102.8 (2020), p. 082002. DOI: 10.1103/PhysRevD.102.082002. arXiv: 2003.06614 [astro-ph.HE].
13. R. Abbasi et al. “Search for dark matter annihilation in the center of the Earth with 8 years of IceCube data”. In: *PoS ICRC2019* (2020), p. 541. DOI: 10.22323/1.358.0541. arXiv: 1908.07255 [astro-ph.HE].
12. D. García-Fernández et al. “NuRadioMC - Simulation Code for the Next Generation of Radio Neutrino Detectors”. In: *PoS ICRC2019* (2020), p. 896. DOI: 10.22323/1.358.0896.
11. P. Allison et al. “Constraints on the diffuse flux of ultrahigh energy neutrinos from four years of Askaryan Radio Array data in two stations”. In: *Phys. Rev. D* 102.4 (2020), p. 043021. DOI: 10.1103/PhysRevD.102.043021. arXiv: 1912.00987 [astro-ph.HE].
10. M. G. Aartsen et al. “Neutrino astronomy with the next generation IceCube Neutrino Observatory”. In: (Nov. 2019). arXiv: 1911.02561 [astro-ph.HE].
9. P. Allison et al. “Long-baseline horizontal radio-frequency transmission through polar ice”. In: *JCAP* 12 (2020), p. 009. DOI: 10.1088/1475-7516/2020/12/009. arXiv: 1908.10689 [astro-ph.IM].
8. J. A. Aguilar et al. “The Next-Generation Radio Neutrino Observatory – Multi-Messenger Neutrino Astrophysics at Extreme Energies”. In: (July 2019). arXiv: 1907.12526 [astro-ph.HE].
7. A. Connolly et al. “Recent Results from The Askaryan Radio Array”. In: *PoS ICRC2019* (2021), p. 858. DOI: 10.22323/1.358.0858. arXiv: 1907.11125 [astro-ph.HE].
6. C. Glaser et al. “NuRadioMC: Simulating the radio emission of neutrinos from interaction to detector”. In: *Eur. Phys. J. C* 80.2 (2020), p. 77. DOI: 10.1140/epjc/s10052-020-7612-8. arXiv: 1906.01670 [astro-ph.IM].
5. B. A. Clark. “Optimization of a Search for Ultra-High Energy Neutrinos in Four Years of Data of ARA Station 2”. PhD thesis. The Ohio State University, Ohio State U., 2019.
4. P. Allison et al. “Design and performance of an interferometric trigger array for radio detection of high-energy neutrinos”. In: *Nucl. Instrum. Meth. A* 930 (2019), pp. 112–125. DOI: 10.1016/j.nima.2019.01.067. arXiv: 1809.04573 [astro-ph.IM].
3. P. Allison et al. “Observation of Reconstructable Radio Emission Coincident with an X-Class Solar Flare in the Askaryan Radio Array Prototype Station”. In: (July 2018). arXiv: 1807.03335 [astro-ph.HE].
2. P. Allison et al. “Measurement of the real dielectric permittivity ϵ_r of glacial ice”. In: *Astropart. Phys.* 108 (2019), pp. 63–73. DOI: 10.1016/j.astropartphys.2019.01.004. arXiv: 1712.03301 [astro-ph.IM].

1. F. Kislak, B. Clark, M. Beilicke, and H. Krawczynski. “Analyzing the data from X-ray polarimeters with Stokes parameters”. In: *Astropart. Phys.* 68 (2015), pp. 45–51. DOI: 10.1016/j.astropartphys.2015.02.007. arXiv: 1409.6214 [astro-ph.IM].